Synthetic Relativity: Tri-modality and the Fixpoint Semantics of the Logical Observer

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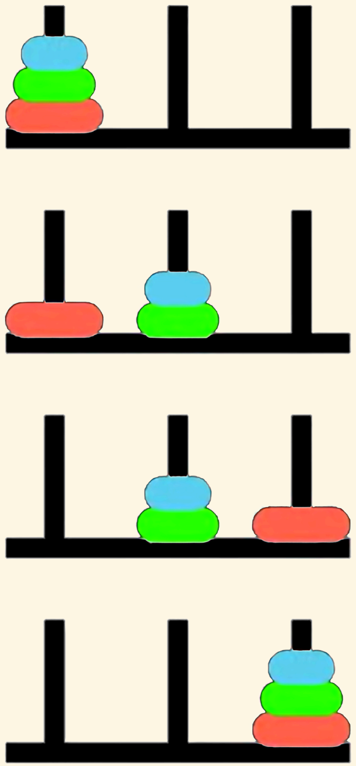
The situation calculus

The situation calculus (Towers of Hanoi)



Fluents: $OnPeg(d, p, s)$

Initial Situation:

$$OnPeg(D_1, Peg_1, S_0) \wedge$$
$$OnPeg(D_2, Peg_1, S_0) \wedge$$
$$OnPeg(D_3, Peg_1, S_0)$$

Actions: $move(d, p_{from}, p_{to})$

Goal: All disks on Peg_3.

Why Towers of Hanoi?

- Naturally highly recursive, problem naturally decomposes into subproblems
- You learn how to solve it during search, for larger instances of n

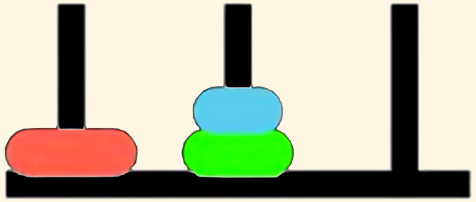
Reiter's Induction Axiom

The Foundational Induction Axiom: Second Order

$$\forall P.\ [P(S_0) \land \forall a, s\,(P(s) \to P(do(a, s)))] \to \forall s\, P(s)$$

- Every valid situation is mathematically guaranteed to be a finite sequence of actions rooted at S_0.
- Establishes the situation domain as an **inductive** datatype.
- Note that standard first order logic cannot enforce reachability due to the Compactness Theorem.

Situations & Actions



Initial situation: S_0

Step 1: $do(move(D_1, Peg_1, Peg_3), S_0)$

Step 2: $do(move(D_2, Peg_1, Peg_2), do(move(D_1, Peg_1, Peg_3), S_0))$

Step 3: $do(move(D_1, C, B), do(move(D_2, Peg_1, Peg_2), do(move(D_1, Peg_1, Peg_3), S_0))$

Step 4-: etc.

Situations as a Least Fixpoint

Let $\mathcal{A}$ be actions and $\mathcal{S}$ be the universe of situations. Define a monotonic operator Γ:

$$\Gamma(X) = \{S_0\} \cup \{do(a, s) \mid a \in \mathcal{A}, s \in X\}$$

Recall that the **Second-Order Induction Axiom** dictates the exact boundaries of the universe:

- **The Premise (If):** $P(S_0) \wedge \forall a, s\, (P(s) \rightarrow P(do(a, s)))$ translates to $\Gamma(P) \subseteq P$. Here, P is a **pre-fixed point** *(an over-approximation that obeys the rules, but may contain unreachable "junk" like ghost situations or disconnected time loops)*.
- **The Conclusion (Then):** $\forall s\, P(s)$ asserts that the true universe of situations $\mathcal{S}$ is a subset of P.
- The axiom combines these, stating $\mathcal{S}$ is a subset of *every* valid pre-fixed point. Taking their intersection mathematically acts as a sieve, filtering out all the unshared, unreachable junk.

Situations as a Least Fixpoint (continued)

The Knaster-Tarski Theorem: least fixpoint

The intersection of all pre-fixed points of a monotonic operator yields its **least fixpoint**.

Therefore, the Second-Order Induction Axiom rigorously enforces

$$\mathcal{S} = \mu X.\, \Gamma(X)$$

(leaving only the pure, minimal valid tree).

Syntactic transformation to First Order Logic

Basic Action Theories [Lin and Reiter 94] define Theory $\mathcal{D} = \Sigma \cup \mathcal{D}_{S_0} \cup \mathcal{D}_{Poss} \cup \mathcal{D}_{SSA} \cup \mathcal{D}_{UNA}$, where:

- $\mathcal{D}_{S_0}$: axioms describing the initial state of the fluents in S_0
- $\mathcal{D}_{poss}$: axioms describing the prerequisites of the primitive actions, of the form:

  $$Poss(A(\vec{x}), s) \equiv \Theta_A(\vec{x}, s),$$

  one for each primitive action A.
- $\mathcal{D}_{ssa}$: axioms describing the effects and non-effects of actions, of the form:

  $$F(\vec{x}, do(a, s)) \equiv \psi_F(\vec{x}, a, s),$$

  one for each fluent F.
- $\mathcal{D}_{una}$: unique name axioms for the actions and objects, e.g. $move(d_1, p_1, p_2) = move(d_2, p_3, p_4) \supset (d_1 = d_2 \wedge p_1 = p_3 \wedge p_2 = p_4)$, $(D_1 \neq D_2) \wedge (D_2 \neq D_3) \wedge (D_1 \neq D_3)$ and $(P_0 \neq P_1) \wedge (P_1 \neq P_2) \wedge (P_0 \neq P_2)$.
- Σ: some foundational domain-independent axioms, e.g. $Smaller(d', d)$

Action Precondition (Possibility) Axioms

$$
\begin{aligned}
Poss(move(d, p_{from}, p_{to}), s) \equiv\ & Disk(d) \wedge (p_{from} \neq p_{to}) \wedge OnPeg(d, p_{from}, s) \wedge \\
& \neg \exists d' \Big(Disk(d') \wedge Smaller(d', d) \wedge \\
& (OnPeg(d', p_{from}, s) \vee OnPeg(d', p_{to}, s)) \Big)
\end{aligned}
$$

A solution to the frame problem (sometimes) [Reiter 91]

1. Positive effect axioms:

$$\exists p_{from}\ (a = move(d, p_{from}, p)) \supset OnPeg(d, p, do(a, s))$$

2. Negative effect axioms:

$$\exists p_{to}\ (a = move(d, p, p_{to})) \supset \neg OnPeg(d, p, do(a, s))$$

3. Frame axioms:

$$OnPeg(d, p, s) \wedge \neg\exists p_{to}\ (a = move(d, p, p_{to})) \supset OnPeg(d, p, do(a, s))$$

4. Completeness assumption (explanation closure):

$$\neg OnPeg(d, p, s) \wedge OnPeg(d, p, do(a, s)) \supset \exists p_{from}\ (a = move(d, p_{from}, p))$$

Successor state axioms $=$ positive effect axioms $+$ negative effect axioms $+$ frame axioms $+$ completeness assumption

Successor State Axioms

Suppose that for fluent $F(\vec{x}, s)$:
$\gamma_F^+(\vec{x}, a, s)$ encodes all *positive* effects; and
$\gamma_F^-(\vec{x}, a, s)$ encodes all *negative* effects.
The successor state axiom solution has the following form:

$$F(\vec{x}, do(a, s)) \equiv \gamma_F^+(\vec{x}, a, s) \lor F(\vec{x}, s) \land \neg\gamma_F^-(\vec{x}, a, s)$$

In our example:

$$\gamma_{OnPeg}^+(d, p, a, s) \stackrel{\text{def}}{=} \exists p_{from}\ (a = move(d, p_{from}, p))$$
$$\gamma_{OnPeg}^-(d, p, a, s) \stackrel{\text{def}}{=} \exists p_{to}\ (a = move(d, p, p_{to}))$$

So successor state axiom is:

$$\begin{aligned} OnPeg(d, p, do(a, s)) \equiv & [\exists p_{from}\ (a = move(d, p_{from}, p))] \lor \\ & OnPeg(d, p, s) \land \neg[\exists p_{to}\ (a = move(d, p, p_{to}))] \end{aligned}$$

Syntactic Transformation: Regression ($\mathcal{R}$)

The **regression operator** ($\mathcal{R}$) evaluates a query about a future state by pushing it backward in time.

It syntactically replaces a fluent evaluated at $do(a, s)$ with the logically equivalent right-hand side of its Successor State Axiom.

$$\mathcal{R}[\boldsymbol{F(\vec{x}, do(a, s))}] \stackrel{\text{def}}{=} \gamma_F^+(\vec{x}, a, s) \vee \boldsymbol{F(\vec{x}, s)} \wedge \neg\gamma_F^-(\vec{x}, a, s)$$

Instead of computing the physical state tree forward (building), the logic engine *rewrites the future query into a question strictly about the preceding state s.*

- By repeating this recursively, any complex historical query is mathematically reduced to a simple database lookup against the initial state (S_0).

Towers of Hanoi Regression Example

Is D_2 on Peg_2 after we move D_1 from Peg_1 to Peg_2?

1. Apply $\mathcal{R}$ to the query:

$$\mathcal{R}[\,OnPeg(D_2, Peg_3, do(move(D_1, Peg_1, Peg_2), S_0))\,]$$

2. Syntactic substitution (using our SSA):

$$\equiv [\exists p_{from}\ (move(D_1, Peg_1, Peg_2) = move(D_2, p_{from}, Peg_3))] \lor$$
$$OnPeg(D_2, Peg_3, S_0) \land \neg[\exists p_{to}\ (move(D_1, Peg_1, Peg_2) = move(D_2, Peg_3, p_{to}))]$$

3. Evaluate using Foundational Axioms ($\mathcal{D}_{una}$): Because our Unique Names Axioms explicitly state $D_1 \neq D_2$, both action equivalences mathematically evaluate to $False$:

$$\equiv \mathbf{False} \lor OnPeg(D_2, Peg_C, S_0) \land \neg\mathbf{False}$$

4. Logical Result:

$$\equiv OnPeg(D_2, Peg_C, S_0)$$

Property Persistence as a Greatest Fixpoint

Property persistence in the situation calculus captures a condition ϕ that holds *now* and is preserved after *any legal action* via regression ($\mathcal{R}$).

[Kelly & Pearce AIJ 2010] define this as the **Persistence operator** ($\mathcal{P}$):

$$\mathcal{P}(Z) \equiv \phi(s) \land \forall a \big(Poss(a, s) \rightarrow \mathcal{R}[\, Z(do(a, s)) \,] \big)$$

Iteratively applying this meta-level calculation computes the maximal invariant without explicitly invoking the second-order induction axiom.

- By wrapping the single-step regression operator inside a monotonic formula transformer ($\mathcal{P}$), the future can be evaluated back in the current state.

The Syntactic Limit: Greatest Fixpoints (νZ)

- Property persistence is an infinite-horizon **safety property**.
- The persistence condition is the **greatest fixpoint** of this operator: $\nu Z.\ \mathcal{P}(Z)$.
- *The Problem:* Regression natively builds finite states. To syntactically evaluate this infinite fixpoint natively, we must shift our mathematical framework.

Category Theory

The Dual Problems: Algebras & Coalgebras

To transition from finite histories to observing infinite Greatest Fixpoints, we harness **Category Theory**. An *endofunctor* (F) over a state space (S) is a structural template that shapes the resulting data object, $F(S)$.

- **Algebra (Situation Calculus / Constructor):** Morphism arrows point *inward*: $F(S) \to S$
  - Our endofunctor shapes the inputs: $F(S) = Action \times S$
  - The *morphism* evaluates to: $(Action \times State_{current}) \to State_{next}$
  - *Concept: Construct* a history from the bottom-up.
- **Coalgebra (Property Persistence / Observer):** Morphism arrows point *outward*: $S \to F(S)$
  - Our endofunctor shapes the emitted data: $F(S) = Observation \times S$
  - The morphism evaluates to: $State_{current} \to (Observation \times State_{next})$
  - *Concept:* Take a running state, break off an *observation*, and unfold the remaining state into the future.

Categorical Proof Techniques: Coinduction & Stone Duality

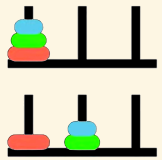
To reason about infinite streams when operating in **coalgebras**, we require two categorical proof mechanisms .

1. Coinduction (The Dual of Induction) While standard induction proves a finite base case and builds upward, coinduction works top-down. It *proposes* a Greatest Fixpoint (an infinite rule, νZ) and verifies it locally. *If the observation holds right now, and the rules guarantee it holds in the next step, the Greatest Fixpoint is guaranteed to hold forever.*

2. Stone Duality (Syntax-Semantics Mirror) In Category Theory, **Stone Duality** dictates a perfect, inverse mirror-image between **Syntax** (logical formulas) and **Semantics** (topological state space).

- *The Syntactic Transformation:* If our logic engine formally declares two distinct observational paths to be identical (Bisimulation Equivalence)…
- *The Semantic Result:* Stone Duality guarantees the underlying semantic space automatically folds (quotients) to make those paths topologically identical.

Example: Coinduction in Towers of Hanoi



We observe the *relative movement* of the smallest disk D_1 (alternating moves with D_2 or D_3) on $OnPeg(d, p, s)$:

$$\Delta p \equiv (p_{to} - p_{from}) \pmod 3$$

1. The Coalgebraic Stream: Solving the puzzle generates an infinite stream of spatial jumps:

$$\langle +2 \rangle, \langle -1 \rangle, \langle -1 \rangle, \langle +2 \rangle \ldots$$

2. The Coinductive Proposal: We propose an idealised Greatest Fixpoint where the disk *always* cycles left:

$$ParityInvariant \equiv \nu Z.\ \langle -1 \rangle Z$$

Example: Coinduction in Towers of Hanoi (Continued)

3. The Coinductive Step (Bisimulation): The physical stream outputs a jump of $+2$. How does our proposed invariant survive this contradiction? To satisfy the local coinductive check, the logic engine natively evaluates the domain's geometry. It formally recognises that jumping right twice ($+2$) is structurally indistinguishable from jumping left once (-1). It applies this coinductive equivalence:

$$\forall \phi. \ \langle +2 \rangle \phi \ \equiv \ \langle -1 \rangle \phi$$

4. The Stone Duality Resolution: By syntactically equating these propositions, **Stone Duality** automatically forces the absolute state space to fold into a modulo-3 *quotient ring* ($\mathbb{Z}_3$)!

This is our new syntactic transformation!

Synthetic relativity axioms

Tri-modality & ternary representation

Logical entailment operates within a **Total Category**—a mathematically complete Stone Duality glues syntax and semantics together.

It is evaluated relative to a specific baseline **Reference Frame**, $\mathrm{S}_{\mathfrak{f}}$. This facilitates the rules and symmetries over which the observer's duality is constructed.

- It utilises a ternary structure, drawing from **Routley-Meyer semantics** for substructural, relevance, and intuitionistic logics, involving relations $R(x, y, \mathrm{S}_{\mathfrak{f}})$.
- This allows us to decouple the synthetic (axiomatic) transformations from the topological space.

Synthetic Relativity Axioms

The space is governed by three interacting modalities.

- RDo **(The Subproblem / Syntax): The Inductive Accumulator ($\Box_1$)** Operates in constructive, non-Boolean logic. Recursively gathers finite, computationally observable properties (compact elements) along a directed logical path.
- ODo **(The Master Problem / Semantics): The Limit/Creator Modality ($\Box_2$)** The semantic evaluator and *interval invariant*. Performs topological closure—evaluating the directed set generated by RDo and snapping it into a discrete, continuous limit.
- PDo **(Autonomous Progression): The Dynamic Transition Functor ($\Box_3$)** The transition modality that governs the forward evolution of the system. Mathematically, it acts as an endofunctor within the Total Category, orchestrating the step-by-step recursive interaction (categorical adjunction) of local syntax (RDo) and global semantics (ODo) relative to S_f.

Constructive Accumulation

Axiom 1: The Axiom of Constructive Accumulation

The Dynamic Transition Functor (PDo) governs the autonomous progression of the system via a strict recursive transition law relative to the Reference Frame (S_f):

- **Base Case (Initialisation):** Let $S_r^{(0)}$ and $S_o^{(0)}$ be the initial states.

$$\Sigma_0 = \mathrm{PDo}\big(S_r^{(0)},\ S_o^{(0)},\ S_f\big)$$

- **Successor Rule (Inductive Step):** For any step $n \geq 0$, the next state strictly nests new inductive syntax (RDo) and semantic coinduction (ODo):

$$\Sigma_{n+1} = \mathrm{PDo}\Big(\mathrm{RDo}\big(R_{n+1}, S_r^{(n)}\big),\ \mathrm{ODo}\big(O_{n+1}, S_o^{(n)}\big),\ S_f\Big)$$

Constructive Accumulation (continued)

Unrolling this recursive law mathematically generates an endlessly nested trace of accumulating evidence:

$$\mathrm{PDo}(S_r, S_o, S_f)$$

$$\mathrm{PDo}\big(\,\mathrm{RDo}(R_1, S_r),\ \mathrm{ODo}(O_1, S_o),\ S_f\,\big)$$

$$\mathrm{PDo}\Big(\,\mathrm{RDo}\big(R_2, \mathrm{RDo}(R_1, S_r)\big),\ \mathrm{ODo}\big(O_2, \mathrm{ODo}(O_1, S_o)\big),\ S_f\,\Big) \ldots \mathit{etc.}$$

It reads: *“Relative to reference frame* S_f*, the accumulation of inductive evidence* RDo *yields the coinductively realised semantic limit* ODo*.”*

Semantic Realisation

Theorem I: Semantic Realisation

Semantic Realisation is the mathematical state where syntax and semantics converge. Because the engine builds upon finite, verifiable observables, autonomous progression halts when the topology reaches Algebraic Compactness:

$$\mu(\mathrm{PDo}) = \mathrm{ODo}\left(\bigsqcup_{i \in R} \mathrm{RDo}_i(\perp)\right)$$

- $\bigsqcup \mathrm{RDo}$ **(Syntax):** The builder generates a raw, directed chain of evidence from the ground up. A syntax string alone has no inherent “meaning.”
- ODo **(Semantics):** The top-down evaluator reads the syntax trace and evaluates its truth value.

Proof Sketch: Corresponds to the computation of the **Kleene Fixed-Point Theorem**.

Semantic Realisation (continued)

- The ultimate stable fixpoint (μ) is caught when ODo evaluates the limit (Supremum, $\bigsqcup$) of the recursively enumerable inductive proofs (RDo).
- This accumulation mathematically begins from an initial state of ignorance ($\perp$), but is constructed and evaluated strictly relative to the rules of the **reference frame** (S_f).
- This mathematically realised limit acts as the interval semantic invariant for that specific frame.

The Synthetic Transformation

Axiom 2: The Axiom of Synthetic Transformation

Once Realisation is achieved, the relativistic reference frame mathematically shifts. The categorically preserved invariant (the semantic limit, O) becomes the new reference frame (S_f) for the next order of progression.

$$\mathrm{PDo}(N, O, S_f) \longrightarrow \mathrm{PDo}(X, Y, O)$$

- **Left Side:** The realised state computed within the initial reference frame (S_f).
- **The Arrow ($\longrightarrow$):** The Synthetic Transformation (Topological closure and categorical Change of Base).
- **Right Side:** A new, higher-order autonomous progression begins. The complex limit O has mathematically synthesised into an atomic, discrete baseline coordinate for the new subproblem (X).

Towers of Hanoi example

1. Local Unrolling (Building the Situation Trace): The engine recursively accumulates primitive actions (moving disks), generating the topological equivalent of a nested $do()$ sequence:

$$\mathrm{PDo}\Big(\mathrm{RDo}\big(\mathbf{move}(D_1, P_2), \ldots S_f\big),\ \mathrm{ODo}\big(O_{\text{target}}, \mathrm{Goal}\big),\ S_f\Big)$$

2. Limit Catching: Through top-down semantic reflection ($\Box_2$), the ODo modality mathematically coinduces a structural property of the optimal traces: The Parity Invariant (O_{parity}).

3. The Synthetic Transformation (Change of Base): The Transformation Axiom fires, executing a categorical Change of Base:

$$\mathrm{PDo}(N_{\text{trace}}, O_{\text{parity}}, S_f) \longrightarrow \mathrm{PDo}(X, Y, \mathbf{O}_{\text{parity}})$$

The invariant (cycle D_1, alternate move, cycle D_1) emerges as the unique, strictly positive proof path through the state space.

- The new parity invariant shifts to become the Reference Frame. By adopting this rule as a **lifted axiom**, *the search tree branching factor strictly collapses from 3 down to 1.*

The Dual Fixpoints

Unifying Reachability (μ) & Persistence (ν): By natively intertwining syntax (RDo) and semantics (ODo), the dynamic transition functor (PDo) mathematically fuses two distinct forms of logical progression:

1. **Induction / Least Fixpoint (μ):** Governed by the RDo modality. The bottom-up synthesis of finite evidence from baseline ignorance ($\perp$). It guarantees computational *Reachability*.
2. **Coinduction / Greatest Fixpoint (ν):** Governed by the ODo modality. The top-down evaluation of continuous semantics. It guarantees *Safety and Property Persistence*.

Theorem II: Algebraic Compactness

Because **Theorem I (Semantic Realisation)** formally computes a categorical *bilimit*, the system achieves algebraic compactness. At convergence, the initial algebra of the dynamic transition functor perfectly coincides with its final coalgebra:

$$\mu(\mathrm{PDo}) \cong \nu(\mathrm{PDo})$$

The active inductive trace (μ) strictly fulfills the persistent semantic invariant (ν).

Theory-Driven Generalization

When the converged limit shifts to become the new Reference Frame (S_f) via the **Transformation Axiom**, it acts natively as a global Greatest Fixpoint (ν) and the coinduced property is added to the reference theory for subsequent computations. The system treats **generalisation** as a categorical operation over its Reference Theory (S_F) facilitating invariant proposal through the foundational duality of fibrations and adjunctions.

- **1. Change of Base (The Fibrational Shift)**
  - Logical properties form a *Total Category* fibered over a *Base Category* of states.
  - The system performs a **Change of Base** from the coinduced space (ODo) to the inductive space (RDo). This functor maps complex, induced histories into coinduced progressions.
- **2. Adjoint Duality (Swapping the Problem)**
  - Generalising a safe invariant is structurally identical to the problem of deductive planning, simply with the functors reversed (an Adjunction: $Left \dashv Right$).
  - *Planning (Left Adjoint):* Inductive, bottom-up synthesis of a finite trace (Least Fixpoint).
  - *Generalising (Right Adjoint):* Coinductive, top-down projection of an infinite bound (Greatest Fixpoint). The finite local observations mathematically push across the adjunction to form the exact coinductive proposal.

Theory-Driven Generalization (continued)

- **3. Universal Properties (Terminal Coalgebras)**
  - The system requires no external engineering templates or “Syntax-Guided” rules.
  - Because S_F natively defines the geometric symmetries of the domain, the coinductive proposal emerges as a **Universal Property** (specifically, a Terminal Coalgebra). Stone Duality ensures that the logical syntax automatically folds to perfectly bound the state space’s innate topology.

Relativistic Domain Theory in Logical Form (R-DTLF)

Domain Theory in Logical Form (Syntax versus Semantics)

To execute our modalities, we require an architecture that natively fuses classical (inductive) logic with intuitionistic (coinductive) logic. We ground our system in Abramsky's **DTLF** (1991), based on an **intuitionistic** construction of truth.

- **Baseline Ignorance ($\perp$):** Computation initiates at Bottom ($\perp$)—the mathematical coordinate of zero information.
- **Compact Elements (Finite Observability):** Basic propositions serve as atomic, observable quanta. In DTLF, logical formulas correspond directly to compact elements, meaning any verified property must be confirmable in finite time.
- **Directed Accumulation ($\sqsubseteq$):** Knowledge grows monotonically. The logic is strictly positive and constructive, meaning once an observation is verified, it establishes an upward, directed path toward the limit without retraction.

The Incompleteness Boundary: Logic remains classically monotonic locally. However, due to relative incompleteness (Gödel's Incompleteness & Beklemishev's reflection principles), a static system cannot resolve global limits. The observer must step outside the local system.

The DTLF Logical Primitives

In DTLF, formulas do not describe static states; they describe observable properties as they unfold. Because a system can only assert what it can finitely verify, the baseline syntax is strictly positive:

$$\phi, \psi ::= \mathbf{t} \mid \mathbf{f} \mid \phi \wedge \psi \mid \phi \vee \psi \mid \text{Constructors}(\phi)$$

- X, Y **(Variables):** Placeholders used to define recursive logical properties.
- $\top$ **(True / Top):** The baseline trivial observation (meaning a valid transition occurred and the system did not crash).
- $\perp$ **(False / Falsum):** The null observation. It represents a paradox, an impossible transition (a crash), or the state of "absolute baseline ignorance" from which new inductive proofs begin.
- $\phi \wedge \psi \;/\; \phi \vee \psi$ **(Conjunction / Disjunction):** "I simultaneously observe both ϕ and ψ" / "I observe either ϕ or ψ."
- $\langle\delta\rangle\phi$ **(The Coalgebraic Transition Modality):** This is the literal "arrow breaking in two." It reads: *"The system makes a visible relativistic jump of δ (where $\delta \in \mathbb{Z}$), and the continuing, unobserved future stream will satisfy property ϕ."*
- $\mu X.\ \Phi(X)$ / $\nu X.\ \Phi(X)$ **(Least & Greatest Fixpoints):** The primitives for recursive behaviour. μ drives finite, bottom-up induction from baseline ignorance (reachability), while ν defines an endlessly repeating, top-down coinductive stream (persistence).

What is missing? Classical Negation ($\neg\phi$) and Implication ($\phi \to \psi$).

- You cannot empirically “observe” an infinite negative.
- If a program runs forever, you can never halt it at a finite step to confidently declare “This will never happen.”
- Negation only emerges at the continuous limit.

During active progression, we must throw out Classical Negation and the Law of Excluded Middle, because as an observer, one cannot empirically observe an infinite negative.

Example: Computing the Hanoi Parity Invariant

We can now express our hypothesis for the Towers of Hanoi natively in DTLF. We want to co-induce an idealised invariant where the smallest disk flawlessly cycles one peg to the right $(+1)$ forever.

1. Proposing the Idealised Rule: We propose this property as our Greatest Fixpoint:

$$ParityInvariant \equiv \nu X.\ \langle -1 \rangle X$$

2. The Reality: However, the physical puzzle generates a stream that forces a forward jump of $+2$ from Peg_1 back to Peg_3. The observable bounds are:

$$Observed \equiv \nu X.\ (\langle -1 \rangle X \vee \langle +2 \rangle X)$$

3. Syntactic Bisimulation Equivalence: Because the topological futures of $+2$ and -1 land on the identical peg, we apply our syntactic equivalence ($\langle +2 \rangle \phi \equiv \langle -1 \rangle \phi$). This cleanly collapses the stream into our idealised property:

$$\nu X.\ (\langle -1 \rangle X \vee \langle +2 \rangle X)\ \equiv\ \nu X.\ \langle -1 \rangle X$$

Spatial Completeness & The Static Limitation

The mathematical power of vanilla DTLF lies in **Spatial Completeness** (rooted in Stone Duality).

- **The Guarantee:** It proves a 1-to-1 isomorphism between the algebraic, step-by-step **Syntax** ($\phi \vdash \psi$) and the continuous, geometric **Semantics** ($[\![\phi]\!] \subseteq [\![\psi]\!]$).
- **Catching Limits:** Computations resolve when the directed path of finite evidence mathematically catches (satisfies) an infinite continuous topological limit (a Supremum).

Vanilla DTLF perfectly describes a single, *static* universe. However, to differentiate induction from coinduction and model dynamic, continuous learning, **the Reference Frame itself must dynamically shift** so we use a relativistic version, termed R-DTLF.

Translating Continuous Syntax into Successor State Axioms

1. Sidestepping the Frame Problem (SSA vs. R-DTLF)

- **SitCalc SSA (Global Tracking):** Evaluates the entire history (s) across rigid Successor State Axioms:

$$On(d, p, do(a, s)) \equiv a = \mathrm{move}(d, p) \vee \big(On(d, p, s) \wedge \neg\exists p'.\, a = \mathrm{move}(d, p')\big)$$

- **R-DTLF (Local Entailment):** Operates purely on verifiable observables. Irrelevant global variables are topologically invisible (no frame axioms are computed during unrolling):

$$\mathrm{RDo}\big(\mathrm{move}(d, p), S_f\big) \vdash On(d, p)$$

2. Endogenous Code Generation (Compiling the Precondition) Once O_{parity} is realised, the engine autonomously exports the invariant into a discrete classical **Action Precondition Axiom (**$Poss$**)** for the new baseline:

$$Poss_{\text{new}}(\mathrm{move}(d, p), s) \equiv Poss_{\text{old}} \wedge \big(\mathrm{d} = \mathrm{D}_1 \leftrightarrow \mathrm{LastMoved(s)} \neq \mathrm{D}_1\big)$$

By autonomously synthesising this SSA, the logic enforces deterministic execution.

Bridging Worlds: Endogenous Logic Compilation

R-DTLF does not discard classical logic; it functionally *generates* it at the semantic limits.

Theorem III: Topological Booleanization (Compiling Successor State Axioms)

Upon semantic realisation, the continuous intuitionistic trace generated by RDo can be losslessly compiled into a discrete, classical First-Order baseline via the **Lawvere-Tierney Double Negation ($\neg\neg$) Topology**.

Proof Sketch: Topos theory formally guarantees that inside every fluid, intuitionistic space exists a rigid Boolean sub-lattice that can be safely projected onto.

- **Compiling Successor State Axioms (SSAs):** This geometric compression is mathematically homologous to Ray Reiter's **Explanation Closure**. The Relativistic Engine acts as an endogenous logic compiler—synthesising complex discovered invariants directly into rigid First-Order logical rules.
- **Hybrid Execution:** Because of this formal theorem, we can securely export newly learned R-DTLF heuristics directly into standard classical Basic Action Theories (BATs). This allows R-DTLF to handle complex structural learning, and legacy solvers (like Golog).

Comparison

Comparison

Feature	Standard Situation Calculus (The "Building" Frame)	Synthetic Relativity (The "Observing" Frame)
Objective	Historical Reachability / Action Execution	Property Persistence / Safety Guarantees
Conceptual Math	Least Fixpoint (LFP)	Greatest Fixpoint (GFP)
Logical Foundation	Second-Order Induction (Algebraic)	Co-induction (Coalgebraic)
Syntactic Transformation	Transformation into First-Order Logic (SSAs)	Transformation to Relativistic Differentials
Computation Engine	Computed as LFP in the Do modality (Golog)	Computed in the R-DTLF logic
Resolution Engine	First-Order Theorem Proving	Bisimulation Equivalence (Quotients)

Synthetic Relativity versus Situation Calculus versus GLB

Aspect	*Synthetic Relativity (R-DTLF)*	*Situation Calculus (BATs/SSAs) $+$ variants of μ-calculus*	*Bimodal Provability logic (GLB)*
Modalities	RDo, ODo & PDo	PDo, $\langle a \rangle \phi$ (possibility), $[a]\phi$ (necessity)	$\Box$, $\Box_\omega$
Ordering	Sequences of finite Posets, non-unique predecessors, non-well ordered	Successor function, Unique predecessors, non-well ordered	Transfinite ordinality: limits, Non-unique predecessors, well-ordered
Completeness?	Spatially complete (continuous geometry) and Turing complete (universal computation via recursive domains); Arithmetically incomplete (triggers Gödel's incompleteness).	Turing complete (can compute anything), but neither spatially complete (no continuous geometry) nor arithmetically complete (subject to Gödel's incompleteness).	Arithmetically complete (flawlessly models the bounds of provability), but neither Turing complete (strictly decidable, cannot compute loops) nor spatially complete.
Other	Algebraic compactness. Kleene's Realizability & BHK (Brouwer-Heyting-Kolmogorov) Interpretation.	Regression Theorem (solves the frame problem). Fixpoint Semantics & Knaster-Tarski Theorem.	Japaridze's Arithmetical Completeness Theorem. Löb's Theorem & Well-founded Kripke Semantics.

- RDo is equivalent to the $\Box$ modality of bimodal provability logic GLB: *"There exists a finite, step-by-step mechanical proof of this property."*
- ODo is equivalent to the $\Box_\omega$ modality of GLB: *"Evaluated at the continuous limit, this property is universally true,"* i.e., this is the modality of *semantic reflection*.